ActInf Textbook Group ~ Cohort 5 ~ Session 14 (Chapter 6, Part 1) ~ 2/5/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 14 (Chapter 6, Part 1) ~ 2/5/2024 | 58:22

all right it's February 5th 2024 we're
in the first discussion on chapter 6 for
cohort 5 so Andrew thank you for
facilitating everyone feel free to raise
your hand and everything and Andrew go
for
it sure
so chapter six we're beginning part two
of the
textbook um this is basically yeah it's
describing a recipe for Designing active
inference models we get as we know from
part one uh specifically chapter 4
especially we've already been introduced
to partially observable markof decision
making processes as a discret Time
version of um kind of one of the the
sort of Exemplar active inference models
that's used um in a variety of
Behavioral experiments as well as the
continuous models um in continuous time
and so this is kind of just getting
into more of like how do we actually
begin like given in part one we've been
introduced to a lot of the main
principles and kind of the framework of
active inference now we're being
introduced to how to actually construct
the models
ourselves um so let's see yep um yeah
the first part we
have the
introduction uh crucial defined like
when designing an active inference model
you want to specify your system of

interest um for me personally it was
easy coming into active inference have
big ideas of what to apply active
inference to um because of course it can
be applied to a variety of living
organis organisms or other systems but
um really whenever you want to design
your own project you do need to specify
a system of Interest what is the
problem um some other thoughts are like
what are the expectations you have of
what kind of Behavioral um like emergent
behaviors you might see or expect to see
versus what ends up happening whenever
you put your model together um so really
uh one one take on it is to really just
start from basic principles like we have
part one of the book we know much more
about something like how active
inference works now start to use that as
a tool for yeah kind of shaping your
your project or your experiment you want
to put together um so you kind of
simplify and grow outwards from
[Music]
there
um section
6.2 there's a very nice brief like um
designing an active inference model in
four steps again and number one
immediately it's which system are we are
we modeling uh in the context of like
statistical marov blankets like how do
we Define the the generative model
versus the generative process like what
what is what is the agent you're trying
to model and what what defines it and
how do you distinguish it from the

environment or the process that's
occurring around it in which it is
operating um what is the most
appropriate form of the model are you
looking at for example some you know
experiment that invol uh involves I
decades um like continuous movements Arc
reflexes those kinds of things in which
case you might be um going for more of a
continuous time model or are you looking
at like are you attempting to um model
uh an animal or a human uh operating in
its environment in which case it might
be um
it might be making discrete choices um
in which case you're looking at more of
a a discrete time model or a more than
likely a
PDP um if the agent's going to be
choosing its actions and has a Time
Horizon and you know you're including
things like expected free energy um not
just variational free energy then step
three how to set up the generative model
choosing what are the appropriate
variables where the priors it's whenever
you kind of get into the you know the
the meat of the of the modeling process
like what are all the specifications
here we need to make in order to
simulate your experiment that you're
going
for and number four again the same thing
but now how do we set up the generative
process because in the same way that you
have to set up the model you also have
to set up the the process in which it's
it's operating kind of intera AC in

with so
6.3 deeper
discussion uh what defines a model what
defines the process where do we draw the
line um what are the the in internal
States the model um what is it inferring
about the process what are what are the
hidden States it's attempting to infer
from the environment around it um
because as we know it has no direct it
has no direct access right to the
environment around it it can only
infer
um get some nice further discussion in
that section thing other things to
consider nested
blankets um how the organism might use
or your agent might use if you're
modeling an agent in an environment in
which it's moving around or acting like
what kinds of objects are in the
environment that it might use as
potential
affordances
um moving on to section
6.4 what is the most appropriate form we
get a little bit more discussion of
things like time scales throughout 6.4
that's a bit more of a lengthy section
with subsections in it you want to
consider things
like temporal depth um they go into more
detail
in
6.4.3 um in which case yeah so for
temporal depth it's like for example you
might have a
PDP where at the lowest level you'll

have for example say you have an agent
in its environment at the lowest level
it's taking in sensory
observations very quick the quickest
Pace the quickest time scale and as it's
doing that th those sensory observations
of course are generating predictions
prediction errors as it interacts with
the upper layers that might be happening
at a at a slower time scale you could
have you know nested levels in which
case like maybe at a higher level the
the agent has various parameters that
are being tuned that're then you know
descending those predictions are
interacting with the ascending signals
based on the incoming sensory
observations so those are all things to
consider
and remembering from chapter five you
can have yeah multiple time scales
operating in the same model some things
happen very quick some things happen
much slower um and they kind of sink
over time
um yeah
it's a lot of the remaining
sections generally get
into yeah much of what I've already
discussed here we start getting into
into the difference between inference
and learning um learning tends to happen
at at slower time
scales um those would be beliefs
about parameters as opposed to Hidden
States so um what what what do you
believe the predictability of a state is
for example it gets into things like

Precision
modulation
um yeah wraps up with talking about more
detail on defining the generative
process and then final section on just a
very brief uh mention of how later in
the textbook it will get into modelbased
Data analysis
so they save going into too much detail
about okay now you've set up the
simulation per chapter six maybe you got
some more additional details on how to
do that in chapter seven or eight
depending on your use case and finally
chapter nine data analysis where you
actually like how do we interpret these
results if you're using um mat lab and
SPM like what what do the and you use a
standard routine which is freely
available from the the software package
that priston at all released like what
do these graphs mean that we're seeing
once we run the simulation um how do we
do individual versus group level
analysis um so yeah um I feel like that
was hopefully a holistic enough starting
point for uh for the discussion
here um oh thank you
Susan um yeah first first time doing
this so happy to be here um yeah does
anyone have
any questions that they're thinking
about anything that's maybe come up
um like or any particular topics or
something they wanted to point to in the
textbook a certain
page I have a question uh can you hear
me yes okay um I wonder if you could

talk a little bit about in the process
of selecting a
model how do you go
about deciding what your mark of blanket
is going to be so
from the modeling perspective can you
just elaborate on the mark of blanket
Choice uh I'm I'm very curious about
that yeah
um so maybe a good quick question um
like do you have an example of
maybe like just any you'd like to look
at a particular phenomena or or
something that you want to understand
better that you might try and make an
experiment to try and study um is there
anything we could start from there
maybe
um I guess I'm I'm too stuck in a
conceptual space right
now and I can't really do that uh but
no sorry let me just say that from a
very abstract
perspective from my question is what is
a
node what is an
edge in your in the models and where
does a mark of blanket come in into the
picture and how do you
decide on the states
that uh
on the states that you choose if I
understand this correctly about Mark
blanket that Define the markco
blanket and if I if I'm incorrect in my
language the way I'm describing these
things let just let me
know sure no I I think the all those

terms are very relevant no edges nodes
we're kind of getting into the Lang of
of graphical models um that's many of
the models that you know this textbook
gets into especially like you know all
these
nice um figures of the PDP for example
those are all you know technically
graphical models um so nodes you know I
don't know if
um I don't know if Daniel it' be
possible maybe to pull up one of the
years in the in the textbook yeah just
mention it and go for it and then maybe
yeah which one yeah um they don't have
any in this chapter but I know that in
the next chapter it gets into pdps so
there might be a good just uh find it
and then put in the chat and then Ali go
for a first pass to magdalen's question
yeah go
for uh yes so thanks for asking the
question because uh I remember asking
the exact same question from Maxwell
ramstead uh about two years ago uh how
can we identify Mark of blankets in
modeling processes and uh I
remember his his answer was um something
to the effect that it's a very
non-trivial process and it depends very
much on uh the situ ituation of interest
and uh on the specificity of um the uh
the kind of modeling or the kind of
questions you try to answer uh in each
modeling scenario uh but uh there was
this paper by Ryan Smith Smith at all I
think it was from 2023 or um 22 I guess
uh towards the end of

2022 uh in which uh they used very
creative approach uh to constructing uh
a kind of markup blanket uh in order to
investigate uh the uh basically uh to
model uh the the the function of the
heart based on um active inference uh
let me pull up the exact title of the
paper but I think it's a very
instructive example as to about how can
we identify uh markup blankets even
creatively uh in every modeling scenario
because I think it's nothing algorithmic
or formulaic about it it requires uh a
bit of out of box out of the box
thinking and creativity in order to uh
to identify suitable uh boundaries in
the uh in the state space uh uh of uh
the mark of blanket you're trying to
um you're trying to construct uh for
each modeling scenario so uh I'll put
the uh Link in the
chat great thank you thank you I I'll
try to give another um way in here so
figure
4.2 shows some motifs in this basian
graph representational format so to your
question about the nodes and the edges
the nodes like the circles those are
variables they could be fixed or they
could be variable and and dynamic or not
and it could be different types could be
a integer it could be a float and so on
this is just very schematic and the
arrows are like causal relationships
that are here directed so then this kind
of maps to X like hidden temperature in
the room observable on the thermometer
here it's like there's two factors that

influence the thermometer readings here
one latent cause and two observables and
here's like something that influences
something that influences and so to
select to to generate structural
hypotheses and to select the different
models and evaluate their differences
like which ones are just like explaining
the most variance but adding more
variables always explains more variants
so then there's these other constraints
that are applied to the model like you
want to explain the most variance
parimon L like basian information
Criterion or you might have a fixed
number of variables to work with or all
these other situational constraints but
because this is a mapping exercise and
Markov blankets are just a feature of
the map like X being a blanketing state
between v and y not carrying any
philosophical implication Beyond just
its insulating role on the graph which
is the map not the territory so there's
a lot of techniques to identify and it
was very wise if Andrew to start with
kind of starting with a specific system
of interest um also the general inquiry
is really interesting but in practice
starting with the specific phenomena or
system is like the starting point on
iterating on something more specific
which helps bring it a little bit more
into the Practical like which variables
you're realistically going to take into
account and which ones are like beyond
the current scope of that modeling
approach

thank you so much and um if you could
elaborate
on what you just said
about the um
the in that on that um figure that you
were just showing if I don't know if you
can put yeah okay so v and y you said
are blanketing
X um can you give me an example from
models that you've worked on or other
people where you know tell me what their
X look like and what the V how the B and
the Y blanketed that
X I'll give one short answer then Andrew
will look forward to your response so
this
Motif is very commonly used to constrain
the structure of time so this is if x is
the present moment this is like saying
the past only influences the future
through the present this is like the
Marian property for time series because
in principle if you had 100 time steps
in a model you could have like time Step
11 influence time step 21 and try to
learn a statistical parameter there but
then you'd end up with so many
parameters so instead under the maravian
simplifying assumption that basically
the present kind of like filters through
like a bottleneck information bottleneck
that simplifies it to only having to
learn one parameter which is like the B
Matrix thank you than helps a
lot yeah and actually Daniel I think
this figure 4.2 is is maybe even a
better um start for versus what I
recommended um just because in addition

to trying to describe the the
specificities related to Markov blankets
maybe just
giving um um kind of an ex like a
concrete example of the the graph in the
upper left corner which the the textbook
describes there um it's so we
have right so we have
um the one impacting with a with an edge
pointing towards the node that is the X
and then uh the two is like so basically
you can see that there's like a causal
relationship here going down like one
impacts X impacts two impacts Y and in
very simplistic terms um so we already
know from earlier in the textbook like a
core part of a generative model is to
have priors which is the one it's our P
of X distribution over hidden States
your beliefs about X then one impacts
what kind of what do you think is going
on right now what is the hidden state
right now that's the X um that you're
inferring or your agent is inferring
your model is inferring um and then that
has a relationship with two which is
your your likelihood Matrix right it's
the probability of the incoming sensory
observations conditioned upon or given
the hidden state that you're Fring so
basically with nodes and edges as Daniel
heavily covered it's really just trying
to figure out like okay what would be
the causal relationships between the
different things going on in your
experiment um and what are those things
in the first place what what are the
particular hidden States um perhaps you

have an example of like um a mouse
navigating am maze um the M the Mouse um
might be pick taking in new sensory
observations for multiple modalities it
might be it can see where it's
navigating it might like feel certain
things it's might might smell certain
things how but those are all things that
you'd want to consider when constructing
a model right so those are all different
variables that end up being their own
nodes and what you end up building out
into what could become a rather complex
graphical model um similarly what are
the states like maybe the mouse is
trying to figure out where it is maybe
it's trying to figure out oh what is the
source of that smell is it a reward is a
well I'm pointing towards this example
in particular because we'll see it in in
h chapter seven I believe um and also
just to just uh I hope you can worry
less knowing that chapter 7 and chapter
eight we'll get into much more concrete
examples of different models as well um
and it'll maybe become a little bit
easier to kind of tie the knot between
all the conceptual considerations and
like an actual experiment you'd want to
set up but that's I'm just trying to
give a little bit more on maybe how you
would construct a model and knowing that
like yeah we do have to understand
graphical models prior to that um it's
just a big web of associations right and
they all lead to a series of
computations and again it's active
inference and so the key part of this

entire process being um a series of computations in order to minimize free energy thank very helpful appreciate that I have a question about the construction of
models um I think my confusion may be that I'm new to this process and it's early on but this is the chapter that um I want to most understand um because I came to active inference with this desire to create a model um for a a longstanding um social science intervention uh my question is um in some when I'm when I first come to a problem you don't always know the variables of interest and so that's part of why you're doing the experiment is that not true in this
case yeah it I mean it to actually any anyone I've had a discussion with who's worked on putting together whether it be active inference models or any other model that might relate to another framework and reinforcement learning or doing other kinds of experiments yeah sometimes it can just just be a very iterative process um more often than not you most people yeah they don't know what all the variables are you might have a strong hunch about what they are or you can of course relate them to like domain knowledge and and secondary literature um but more often than not you may want to construct a model that uses what you know or think is highly likely that this is how this works um have a look at the result result look at what's going on if you're able to to run

the simulation see what the output is
see if it makes any kind of kind of
sense um uh I have a social sciences
background and so things like causal
inference where we're like okay well
what are the variables here uh what are
the confounding variables here what are
we leaving out what are the things that
cannot be observed right so so if you
start seeing like what you might and it
you know what defines an odd observation
is very context specific but if you do
start seeing odd results it might point
to the idea that yeah you might very
well be leaving leaving something out
and then you may want to also double
check the the complexity of your model
like say there's something at the lower
level um that you've defined everything
makes sense it's at this time scale Etc
maybe there's a variable that's left
been left out at another level
um maybe there's something else that's
impacting all of this that's been left
out that that isn't even occurring at
the lower level that you've been focused
on it's been at a higher level this
whole time um a lot of my explanation
there might be might be a little vague
but again yeah it's just very context um
context sensitive I almost want to say
so no thank you
that's I'll add a few more notes on this
um so in the symbolic Active Space and
and an Institute participant JF his Lego
robots using a a logical program rather
than statistical distributions but the
idea is basically the same it's like

rather than a causal um graph which is like we're talking about the textbook it's a causal Theory and here he talks a lot about that incremental proposal so you ask are they actual variables or the hypothesized latent they are actually hypothesized so they're in that Ed in their

hypothesizing which is abductive inference is so generation of hypothesis and then evaluation of hypothesis so that's been analyzed as kind of like itself an active inference process because which hypotheses to propose and then given the hypotheses which actions to take for their to uniquely resolve and learn those are all questions that can be approached in actim and then just one one last technical note is this relatively recent pre-print with friston at all this talks about a statistical problem called structure learning so here it's in the context of the digit um mnist data set this is a very familiar problem and data set to um look across different handwriting Styles and identify the digits and this work which we could probably discuss more in a future time goes into a lot of detail with the structure learning which is like when you have um you have 11 handwriting Styles should you propose an 11th cluster and then you evaluate whether the 11 cluster model or the 10 cluster model is better explaining the data it's like doing better on that parto optimal front and then like 11 is accepted and

then like there's a proposal to
basically to increase the number or
decrease the number of latent factors
and then there's a Criterion for like
accepting a rejecting those kinds of
model updates and so that's a capacity
that's existed for statistical models
outside of actim to kind of like
adaptively dial up or dial down the
dimensionality but it's not a totally
solved problem and it's not totally
implemented in ACM but in principle um
like a system wouldn't need to know the
dimensionality of its environment
because it might just have a and it
doesn't ever need to like it could just
like too cold just right too warm and
then that doesn't mean that that that it
ever needs to like learn the richness of
the continuous distribution of the
temperature in the room yeah that might
be good enough for the air conditioner
just to have three states or something
right right right right great thank you
so
much
I guess I did want to add um there's a
nice
paper um for anyone who's more
interested in like so social science
research or something along those lines
has been a very nice
um series of papers put out by a
researcher um Ryan excuse me Ryan Smith
um he's actually we have several live
streams of him uh giving a tutorial on
active inference with the Institute um
and specifically how to construct models

in in mat lab for anyone who's who's familiar with that or interested um and I bring that up because he has a this paper um on simulating uh what's called exposure therapy so this is research that's that is intended to um Aid interventions um for psychotherapists like can we simulate for example um a client um who um who is suffering from you know some kind of uh avoidance or anxious um disorder that's stimulated by something say they have a very strong phobia of something um computationally it could be anything it could be a spider it could be you know seeing the car accident a variety of things and I so bringing it up like they had to construct a model of like how do you simulate exposure therapy so we already know we have a client um that we're trying to simulate we want to simulate them going through exposure therapy being exposed to the thing that produces the fear in them um to see like you know this is a safe environment that we're simulating in order to like test things out as opposed to Jumping immediately to Empirical research that you know we there too many ethical considerations to just playing around with showing people things that that trigger them um so when constructing this model they like you have to model here's the agent you have to figure out what defines the agent um in their case the agent uh experiences affect or something like emotion um they compose that as a

distribution like what are the emotions
they keep it very simple they go with
something in
Psychology uh referred to as veilance so
you have a nice like positive negative
scale um there's been a lot of nice work
on emotion in general from an active
inference perspective that yeah um
veilance seems to hold up pretty well in
certain context so they go with that um
and then you have um you know how does
the the agent deem the stimulus that
they're being shown in this case let's
say the the agent has a fear of the
spider um so they might interpret it as
dangerous as opposed to safe so now
again we have another kind of sliding
scale um these are all what could be
viewed is like continuous distributions
right like they can be viewed as a
number that moves into the positive the
negative um then you can take those
kinds of things view them as variables
then put them in the model um in this
case we would hope that the agent being
exposed to like say a spider that
ultimately is safe um again bear with me
that it might come off as a silly
example but um say it's a fake spider
this is exposure therapy
you know they're not going to show them
real spider um but to get the agent to
slowly become more comfortable with
being around the spider like we want
those parameters those beliefs about how
dangerous the spider is to change over
time so that also means you have to
we're defining the generative process

the spider's there it's a safe spider
the agent is interacting with it over
time as we run more trials we can see
how do the AG
beliefs begin to change do they change
rapidly does the situation become worse
are they being exposed to The Spider and
it's just freaking them out more they
become they're concerned that it's even
more dangerous um and I'll quickly I
realize I haven't shared the link for
that paper so I've just shared that um
anyway so without going into too much
more detail because it takes some time
um but you know if this paper interests
you go for it
um the whole idea here yeah is to to
just you know thinking through what what
is a variable what is considered as a
variable and a lot of it centers around
defining your agent defining the
environment or process around them and
then trying to find the linkage between
you know all the conceptual ideas like I
said earlier like affect is that there's
plenty of domain knowledge on that in
psychology right like you how do you
make a one: one ratio between a little
node or variable and a graph with you
know research that's been done by your
you know the field you're working from
or interest in like in Psychology how do
we make those connections so that we can
build out a model from there
um yeah and so yeah Daniel's pulled it
up there you know there are different
ways that people attempt to show their
models um this one the notation is

certainly different from the textbook
but I think it's more explanatory right
like we have an agent is being exposed
to a safe stimulus or a dangerous
stimulus like a like a spider we have
their beliefs about its safety or how
danger it is um they also have an
arousal level like you know you can
include things like you know with
psychological research and things like
that we care a lot about physiological
signals is is the agents like heart rate
increasing which by the way you'd want
to consider that probably to be a
continuous variable right um because
it's this is happening very quickly over
time um whereas belief context safe
versus dangerous that could be viewed as
discreet we have two options there um so
those those are also big concerns like
you figure out your variables are they
discreet are they continuous do they
happen at quick time scales Moment by
moment like the agent very quickly is
feing feeling their their heart rate
change um and yet over time as we run
more trials with more exposure therapy
sessions um at a slower time scale their
beliefs start to change right to ideally
a situation where we can find maybe use
this model to find some insight into how
to improve intervention for for uh
people suffering from phobias or anxiety
that's generated by a particular
stimulus um so yeah that's I hope that
that provides some more insight into
yeah how to take a lot of this abstract
or conceptual material and then relate

it more directly to concrete you
situations and problems of Interest
Daniel thanks this is a great paper to
show and I think the figure here shows
like this is like the semantic level
like not worrying about how big the
variable is and so it's kind of like a
little template that could be adopted
just thinking about this situation of
the exposure and then what is the thing
that one is modeling and then what other
phenomena they do they need to further
include like planning is not here and
then the bottom is actually the shape of
the variables thems the
a the b c and d this agent just like the
pdps in the generative model and
described in model stream one and these
are the actual shapes of the variables
and even their types ranging from this
continuous belief to 2 by tws affordance
matrices with context preferences what a
flat preference looks like versus a
sharp
preference it's all kind of
syntactically displayed in this
um flat
L does anyone else have any
other thoughts or things that you know
they're kind of pondering right now uh
regarding this
chapter
do we have any other questions that have
already been maybe asked in
theota
what do you think are good small
experiments to do between the first and
the second discussion on chapter 6 given

what anyone has experienced or
tried we could also look at a
notebook like look at an executable code
right now like a PDP
example or we could talk about
what um else could help supplement
six
I I think let's look at the PM DP
example if it's
okay okay I'll put the link in the chat
but here's pmdp
documentation so this is the python
package by Connor Hines Alex shance daph
Deus
Etc um there are several tutorials and
Demos in this open- Source
package
um the simpler one are tutorial one and
two then the t- maze is most like the
textbook and the epistemic chaining
demonstrates it's kind of like a tze but
with like a little bit more of an
epistemic or slightly different
epistemic
components and each of these
tutorials very nicely provided by the
authors there's like a collab
notebook so just going to this and you
can you can make a copy right from
there um I'll kind of give it back to
Andrew to continue and then maybe next
week we could prepare to execute
something just so that we're not like
needing to execute on the fly but
suffice to say like these notebooks do
work and you see a similar specification
AB c
d That's what defines the

PDP those variables they're
representable in terms of this grayscale
heat map
visualization sometimes um black or
white is one or zero like you just have
to look at the the the um graphical
settings but they describe um how the
variables are
shaped and and so it can help to just
see like okay here's block by block
here's all the variables that need to be
defined here's what the different slices
in the variables
are um yes copying and pasting images or
screenshots or code blocks into language
models is super helpful as they're very
Adept at this kind of
classical basic code exp uh
explanation and these notebooks just go
through Define a b c
let's see if we can collapse and just
look at the sections D so a mapping
between hidden States and observations
ambiguity Matrix B transition Matrix for
the hidden States that's how the hidden
states are inferred to change through
time in this action dependent way and
then policy selections like selecting
which slice from B is going to be used
in the time Evolution C preference over
observations D the prior on the hidden
state which kind of kicks off the chain
but after d is used once then it's just
B and the hid State s unfolding through
time so then to specify the agent it's
just taking in these exact four
variables so adapting active inference
in the pmdp setting to like um whatever

it is that you're trying to model it
does come down to making the right shape
a b C and D it's specified like this
here is a class object for the game
itself so here's like where the rules of
the game like the slot machine or
whatever it turns out those um like or
if it's prisoners dilemma those are
parameterized in something like this so
it's like we Define the whole agent
generative model then the the context
the environment that it kind of plays in
and then here's the active inference
Loop
and here's the pseudo code
basically for that UDA Loop like observe
Orient decide act it's basically like
that kind of pseudo
code invoking the specific game and the
agent and the
interface so this is not like what high
throughput production actm generative
models look like but each of these
tutorials are super informative because
they do work and they are um um forkable
and they do like capture all that's
needed and so that can help demystify it
a lot it's not like an infinite
parameter fishing investigation it's
like build ABCD underscore one and then
try underscore two like a variant and
just try recombining them and it's about
building like a portfolio of different
abs and C's and D and agents and
visualizations and Diagnostics and
summary statistics for the
simulation and these notebooks go a long
long way and until you're using like

large data sets or anything like that
poab will have enough computational
resources
Susan thank you wonderful um
so guess this um you know in terms
of defining an agent um and this kind of
is in somewhat inspired by Cheryl's
question um in terms of goals roles and
context
and you what I find rather elusive and
intriguing
um is you know is the um overall
objective of gen you whether that be
generative modeling or the generative
process is is are we are we seeking
certainty or are we seeking
insight and and and so in terms of you
how I perceive and and quite frankly I'm
going to you know do a plug for my
Thursday at 11:00 a.m. um uh foraging
Insight because what I'm trying to do is
is really focus on how how people
self-model versus how we're trying to um
predict how people are going to react
which is different than you know trying
to program something so although you
know definitely we want to be able to
use technology to augment
agency this comes down to how do we help
people um discover their own agency and
affordances within certain
parameters um you know how do we how do
we translate that
into understanding the
textbook does that make sense to
anyone
I really like that
distinction Insight versus certainty

I've never heard I've never heard that
made that way before but that's so clear
thank you so much um so I can see a real
problem when you
enter enter you know an an experiment
thinking that you're looking for
certainty and you're actually looking
for insight I mean I can I can see how
one could confuse those things when
they're not
distinct yeah like the whole risk
management
industry but to kind of go to equation
2.6 and then pros this really does map
to when we're selecting which experiment
to design policy to undertake the
certainty is is finding observations
that align with our expectations
preferences sampling the homeostatic
body temperature like pursuing certainty
about our homeostasis for temperature
and then literally the Insight or
Information Gain and so and then this
Ties on this is a big topic so we won't
go into right now but different um
scientific epistemologies like does
science prove does it disprove oh well
they confirmed this in an experiment so
they found what they were looking for or
they were shocked by this so then what
did they expect and then our experiments
for falsifying or confirming
or
neither and how we hold on to beliefs
that may not be um you
prudent but you know why we why that's a
surprise
versus um you being able to belief

update so you know this really calls
into question yeah what is learning I
mean you that might be a good good start
is is what is learning
because is learning consuming
information or is it digesting it and
taking action on
it and learning from the action I me
I'll I'll borrow from something Daniel
said earlier as is and you might have to
reiterate it Daniel but uh um you moving
the Pebbles moving the Pebbles that's
what matters
and until we actually act on the world
we're not moving the pedals we're just
new we're just uh regurgitating and and
uh F
into uh uh uh
reminition
wow
wow I actually just shared
um I personally coming from like a
comput computational Psychiatry um sort
of background SL that's my current field
of interest and uh I just shared this
actually since you mentioned the word
rumination there's another paper
Elsewhere on um attempting to simulate
uh simulate rumination as a kind of
cognitive process that can arise based
on certain circumstances I I shared this
one in particular because it it's it's
one of the papers that I've seen so far
that best tries to
describe um in a more kind of human
relatable view uh like what it is to
self-model in one's environment because
as we know like one of the bases of

active inference is free energy
principle um an agent needs to have an
understanding of what separates itself
from its environment in order to
maintain itself otherwise it kind of you
know however we want to phrase it from a
physics point of view it's you know the
environment is entropy it's it's you
dissipate if you just you know give to
everything that's around you you have to
breathe in oxygen you have to do a
variety of things that maintain what is
you versus what is outside of you and so
yeah it's it's really crucial um I like
this paper because it it describes like
a little bit more in depth like for a
human being what what might be required
or involved in self-modeling how
important it is to have a model of
oneself um the role of things like
sensory attenuation which is another big
aspect of active inference like in order
to move your arm you need to like be
able to briefly put simply you need to
forget that your arm is one spot you
need to believe that you can move it to
the place you want to move it to so what
you're doing is you're attenuating or
kind of ignoring where it is right now
you're in motion you're trying to be in
motion right so if you overly believe
like you keep that that belief fixed
that your arm is in one spot it's going
to stay there you have to move it um
that becomes really important when
self-modeling too in order to operate in
the bigger world around you um so so
just all these f Minor Details and

aspects of self modeling I think it's a
fascinating topic though for sure so is
is that the paper that um uh Andrea
shared is that the
overthinking is that covered in that
okay yeah yeah that's thanks for sharing
yeah sure it doesn't get into the IT
specific topic of interest is
depersonalization disorder but I think
the first half of it is really Broad and
I just you know I think it's a nice
introduction to just thinking about
self-modeling in general regardless if
you're interested in Psychiatry or not
cool yeah thank
you pros and then Andrew you can kind of
end it however you want uh yeah Daniel I
have a basic question like I was
wondering the pmdp which you showed uh
can it also handle hierarchical
models yeah it
can okay I don't know if there's a
tutorial on it but we can ask the
authors but we can look through
literature and so on but yes of it
can okay I was okay thank you thank
you I also mention that um the um Daniel
just put in the coda there I shared the
Ryan Smith tutorials um and while those
use Matt lab um a lot of the pmdp stuff
is the coding it's based upon um a lot
of what goes on uh like basically active
inference modeling It generally started
with mat lab and SPM which is what Ry
Ryan Smith and everyone else covers and
then Connor Hines a variety of other
researchers put together pmdp much of it
being based on um the prior work in that

lab and so at least for a verbal
explanation of hierarchical modeling uh
I believe it's either the second or the
third video of the series with Ryan's
Smith um if you so I guess that would be
like model stream 00
1.3 versus the one that's on the coda
there 00 1.1 um just if you want like a
more verbal explanation of kind of how
that that can work in a more direct
coding setting um as a
programmer yeah just wanted to mention
that and um yeah um I don't know I think
we're just about out of time here
but um just brief recap this was chapter
six or recipe for putting together
models uh potentially it it provokes
more questions than it answers um but
it's it I think it gives a frame it
basically brings up all the questions
that you need to be concerned with
whenever you're putting together a model
right what is the agent what defines the
process what are the variables are they
discret are they continuous how do they
work together how do we relate them to
domain
knowledge um and and actually translate
between you know the the language of
basing statistics and mechanics with uh
you if you're a social scientist or or
um an engineer um and
so yeah kicks off chap uh part two of
the book overall the next chapter we'll
get into
continuous models and I think that's it
might be the case that for a good deal
of people here who are interested in

modeling it might be the case that
you'll be looking at a
PDP um like that's what's used most
frequently with trying to model like
human beings in their environments um
animals as well but they're used um they
they involve including things like
planning a Time Horizon uh they don't
have to but um they can and um yeah
that's I hopefully it'll give you much
more concrete examples of thinking
through like how do we translate between
a model and what we're actually trying
to do here uh hopefully chapter 7even
basically will clear up some questions
you might have had from from chapter six
so thank you awesome work Andrew thank
you everybody see you next time thanks
everyone bye bye